

Source code - Secure polling booth using RFID and live vote tracking

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <SoftwareSerial.h>

#include <Keypad.h>

#include <avr/pgmspace.h>


// --- GSM setup ---

#define GSM_RX A0

#define GSM_TX A1

SoftwareSerial GSM(GSM_RX, GSM_TX);


// LCD setup

LiquidCrystal_I2C lcd(0x27, 16, 2);


// Keypad setup

const byte ROWS = 4;

const byte COLS = 3;

char keys[ROWS][COLS] = {

    {'1','2','3'},

    {'4','5','6'},

    {'7','8','9'},

    {'*','0','#'}

};

byte rowPins[ROWS] = {8, 7, 6, 5};

byte colPins[COLS] = {4, 3, 2};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);
```

// Stored UUIDs

```
const char storedUID[10][11] = {  
    "530263E5FA","4700320C11","530263A233","4900303601",  
    "4A00CB63D4","5302639EFA","4A00CB632B","43008A49F2",  
    "430089A796","430089F7C8"  
};
```

// Passwords

```
const char passwords[10][5] = {  
    "123","234","345","456","567","678","789","890","901","012"  
};
```

// User names in PROGMEM

```
const char userNames[10][16] PROGMEM = {  
    "Praisyl Angel","Muthu selvi","Santhiya","Nirmala",  
    "Karthik","Anita","Ravi","Latha","Arun","Priya"  
};
```

// Phone numbers

```
const char phoneNumbers[10][14] = {  
    "+917845330928","+919790474372","+919361078221","+919150366497",  
    "+917845330928","+919790474372","+919361078221","+919150366497",  
    "+919361078221","+919790474372"  
};
```

// Voting status

```
bool voted[10] = {false,false,false,false,false,false,false,false,false,false};
```

```
char receivedID[14];
```

```
char enteredPassword[5];
```

```
int passwordIndex = 0;
```

```
int currentUser = -1;
```

```
// Candidate pins & votes
```

```
const int Candidate1=9, Candidate2=10, Candidate3=11, Candidate4=12;
```

```
int VoteCount1=0, VoteCount2=0, VoteCount3=0, VoteCount4=0;
```

```
// --- Function to read name from PROGMEM ---
```

```
void getUsername(int index, char* buffer) {
```

```
    for (int i = 0; i < 16; i++) {
```

```
        buffer[i] = pgm_read_byte(&(userNames[index][i]));
```

```
        if (buffer[i] == '\0') break;
```

```
    }
```

```
    buffer[15] = '\0';
```

```
}
```

```
// --- GSM SMS function ---
```

```
void sendSMS(const char* number, const char* message){
```

```
    GSM.println("AT+CMGF=1"); // Text mode
```

```
    delay(500);
```

```
    GSM.print("AT+CMGS=\"");
```

```
    GSM.print(number);
```

```
    GSM.println("\");
```

```
    delay(500);
    GSM.print(message);
    delay(500);
    GSM.write(26); // CTRL+Z
    delay(2000);
}

void setup() {
    pinMode(Candidate1, INPUT);
    pinMode(Candidate2, INPUT);
    pinMode(Candidate3, INPUT);
    pinMode(Candidate4, INPUT);

    Serial.begin(9600);
    GSM.begin(9600);

    lcd.init();
    lcd.backlight();
    lcd.setCursor(0,0);
    lcd.print("Place RFID Card");

    // Test GSM
    GSM.println("AT");
    delay(1000);
    GSM.println("AT+CMGF=1");
    delay(1000);
}
```

```

void loop() {
  // --- RFID detection ---
  if(Serial.available()>=14){
    for(int i=0;i<14;i++) receivedID[i]=Serial.read();

    char uid[11];
    for(int i=1;i<11;i++) uid[i-1]=receivedID[i];
    uid[10]='\0';

    currentUser=-1;
    for(int k=0;k<10;k++){
      if(strcmp(uid, storedUID[k])==0){ currentUser=k; break; }
    }

    while(Serial.available()) Serial.read();

    lcd.clear();
    if(currentUser==-1){
      lcd.print("Wrong ID");
      delay(2000);
      lcd.clear(); lcd.print("Place RFID Card");
    } else {
      if(voted[currentUser]){
        lcd.print("Already Voted!");
        delay(2000);
        lcd.clear(); lcd.print("Place RFID Card");
      }
    }
  }
}

```

```

    } else {
        lcd.print("Hello,");
        lcd.setCursor(0,1);
        char nameBuffer[16];
        getUserName(currentUser, nameBuffer);
        lcd.print(nameBuffer);
        delay(2000);
        lcd.clear();
        lcd.print("Enter Password:");
        passwordIndex=0;
        enteredPassword[0]='\0';
    }
}

// --- Password entry ---
if(currentUser!=-1 && !voted[currentUser]){
    char key = keypad.getKey();
    if(key){
        if(key=='*'){ // Check password
            lcd.clear();
            if(strcmp(enteredPassword,passwords[currentUser])==0){
                lcd.print("Vote Accepted!");
                delay(1000);
                lcd.clear();
                lcd.print("Press a Vote");
            }
        }
    }
}

```

```
while(true){  
    if(digitalRead(Candidate1)==HIGH){  
        VoteCount1++;  
        lcd.clear(); lcd.print("Voted A");  
        sendSMS(phoneNumbers[currentUser],"Your vote: Candidate A");  
        delay(2000); break;  
    }  
    if(digitalRead(Candidate2)==HIGH){  
        VoteCount2++;  
        lcd.clear(); lcd.print("Voted B");  
        sendSMS(phoneNumbers[currentUser],"Your vote: Candidate B");  
        delay(2000); break;  
    }  
    if(digitalRead(Candidate3)==HIGH){  
        VoteCount3++;  
        lcd.clear(); lcd.print("Voted C");  
        sendSMS(phoneNumbers[currentUser],"Your vote: Candidate C");  
        delay(2000); break;  
    }  
    if(digitalRead(Candidate4)==HIGH){  
        VoteCount4++;  
        lcd.clear(); lcd.print("Voted D");  
        sendSMS(phoneNumbers[currentUser],"Your vote: Candidate D");  
        delay(2000); break;  
    }  
}
```

```

        voted[currentUser]=true;
    } else {
        lcd.print("Wrong Password!");
        delay(1500);
    }

    // Print vote counts
    Serial.print("A"); Serial.print(VoteCount1);
    Serial.print("B"); Serial.print(VoteCount2);
    Serial.print("C"); Serial.print(VoteCount3);
    Serial.print("D"); Serial.print(VoteCount4);
    Serial.print("E");
    Serial.println(VoteCount1+VoteCount2+VoteCount3+VoteCount4);

    lcd.clear(); lcd.print("Place RFID Card");
    currentUser=-1;
} else {
    if(passwordIndex<4){
        enteredPassword[passwordIndex++]=key;
        enteredPassword[passwordIndex]='\0';
        lcd.setCursor(0,1); lcd.print(enteredPassword);
    }
}
}
}
}
}

```